

Question number	Answer	Additional guidance	Mark
5(a)	• $3.2 \times 10^7 / 3.18 \times 10^7$ (1)		(1)

Question number	Answer	Mark
5(b)(i)	The only correct answer is A <i>B is incorrect because glycosidic bonds are in carbohydrate molecules</i> <i>C is incorrect because hydrogen bonds are between water molecules</i> <i>D is incorrect because there are no ionic bonds in water</i>	(1)

Question number	Answer	Mark
5(b)(ii)	The only correct answer is B <i>A is incorrect because ATP does not store hydrogen</i> <i>C is incorrect because plants and seaweeds do not store glycogen</i> <i>D is incorrect because RUBISCO is not a fuel</i>	(1)

Question number	Answer	Mark
5(b)(iii)	The only correct answer is C <i>A is incorrect because carbon dioxide and hydrogen combine during the light-independent reactions and oxygen is released during the light-dependent reactions</i> <i>B is incorrect because carbon dioxide and hydrogen combine during the light-independent reactions and oxygen is released during the light-dependent reactions</i> <i>D is incorrect because carbon dioxide and hydrogen combine during the light-independent reactions and oxygen is released during the light-dependent reactions</i>	(1)

Question number	Answer	Additional guidance	Marks
(c)	Indicative content: Diagram 1: • seaweed taking up space of other organisms • {fish / sea mammals} could get caught up in wires • could prevent {fishing / water sports / water activities / shipping} Diagram 2: • seaweed could reduce {food / habitat} of water animals • seaweed could compete with other (wild) aquatic plants for light • could provide labour for local population • increasing the economy of the area • increase in carbon dioxide removed from {atmosphere / water} / reduces greenhouse effect • machinery used to {grow / harvest} seaweed might disturb other organisms • causing them to {move away / be stressed} • interrupting food chain • presence of seaweed might provide {food source / habitat} for different species • which would compete with current species • interrupting food chain • increase biodiversity • machinery used to {transport / process} seaweed will burn fossil fuels Other ideas: • maybe other organisms brought into area on the seaweed • which would compete with other species for {food / habitat} • could bring in diseases • that could weaken or kill other species • will Europe have high enough {temperature / light levels} to sustain growth	Level 1: Simple comments about aspects 1 mark = one relevant comment 2 marks = simple comments relating to at least three aspects Level 2: discussion about aspects linked to at least one diagram / idea 3 marks = at least one aspect discussed in more detail 4 marks = at least two aspects discussed in more detail Level 3: detailed discussion about aspects from both diagrams 5 marks = at least three aspects discussed in more detail 6 marks = at least four aspects discussed in more detail which includes an idea not included in the diagram	(6)

